NEPAL COLLEGE OF INFORMATION TECHNOLOGY

Level: Bachelor	Assessment	Year	: 2026
Programme: BE		Full Marks	: 100
Course: Natural Language Processing		Pass Marks	: 45
		Time	: 3hrs.

Candidates are required to give their answers as far as practicable in their own words.
The figures in the margin indicate full marks.
Attempt all the questions.

1. **a)** What is the Turing Test? Explain its relevance to Natural Language Processing and how it relates to the concept of human language and intelligence. **[7]**
   b) Compare NLU and NLG. Also Explain different types of ambiguity in NLP with examples. **[8]**

2. **a)** What is tokenization? Explain one hot encoding method with example and its major drawback. **[7]**
   b) Explain stemming and lemmatization. Compare their differences with examples and discuss when to use each technique. **[8]**

3. **a)** What is TF-IDF? Calculate the TF-IDF scores for each unique term in the following documents (use raw TF and log base 10 for IDF): **[7]**
   - **D1:** "machine learning is a subset of artificial intelligence"
   - **D2:** "deep learning is a subset of machine learning"
   - **D3:** "artificial neural networks enable deep learning"

   b) What is Zipf's Law? How does it relate to term frequency in text documents? Explain with an example. **[8]**

4. **a)** How do word embeddings capture semantic relationships? Explain with the classic example of 'king - man + woman = queen'. **[7]**
   b) What are the limitations of traditional word embeddings like Word2Vec? How do contextual embeddings (like BERT) address these limitations? **[8]**

5. **a)** What is a language model? Explain unigram, bigram, and trigram language models with examples. **[7]**
   b) Using the Naive Bayes algorithm, predict the class of the test document based on the training documents provided: **[8]**
   - **Training Class A:** "abac", "baabaaa", "abbaa"
   - **Training Class B:** "bbaabbab", "abbb", "bbbaab"
   - **Test:** "aabc"

6. **a)** Explain the self-attention mechanism and multi-head attention in Transformers. Why are they important for capturing long-range dependencies? **[7]**
   b) Compare the Transformer architecture with RNN/LSTM architectures. What advantages do Transformers have for sequence modeling? **[8]**

7. **Write a short note on: (Any two)** **[10]**
   - **a)** POS Tagging
   - **b)** Next Sentence Prediction
   - **c)** Cosine Similarity

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